



# Arithmetic sequences

---

1 Match the keyword to its definition. Write the correct letter in each box

|   |                     |
|---|---------------------|
| a | arithmetic sequence |
| b | common difference   |
| c | initial term        |
| d | successive terms    |

|  |                                                         |
|--|---------------------------------------------------------|
|  | a sequence that increases or decreases by the same step |
|  | the constant additive rule from one term to the next    |
|  | terms that are next to each other in a sequence         |
|  | the first term in a sequence                            |

2 Match the arithmetic sequences with their descriptions. Write the correct letter in each box

|   |                             |
|---|-----------------------------|
| a | 7, 5, 3, 1, ...             |
| b | 7, 9, 11, 13, ...           |
| c | -1.6, -1.3, -1, -0.7, ...   |
| d | -1.6, -1.9, -2.2, -2.5, ... |
| e | -2, 3, 8, 13, ...           |
| f | 38, 33, 28, 23, ...         |

|  |                                                  |
|--|--------------------------------------------------|
|  | initial term 7 and common difference $-2$        |
|  | initial term $-1.6$ and common difference $-0.3$ |
|  | initial term $-1.6$ and common difference $0.3$  |
|  | initial term 7 and common difference 2           |
|  | ninth term $-2$ and common difference $-5$       |
|  | fifth term 18 and common difference 5            |



3 Which of these sequences are arithmetic? Tick 2 correct answers

☐ 7, 2, -3, -8, ...

☐ 4, 1, -1, -4, ...

☐ 17, 24, 32, 39, ...

☐  $\frac{4}{5}, \frac{1}{2}, \frac{1}{5}, -\frac{1}{10}, \dots$

☐ -1, -2, -4, -8, ...

4 In which of these sequences do the number of shaded squares in each term form a linear sequence? Tick 2 correct answers


☐
☐
☐
☐

5 An arithmetic sequence has 3rd term = 41 and 7th term = 65. What is the initial term?  
Fill in the blank

\_\_\_\_\_

6 Which of these sequences are arithmetic? Tick 4 correct answers

☐ a, 2a, 3a, 4a, ...

☐ a + 5, a + 8, a + 11, a + 14, ...

☐ a - 2b, a + b, a + 4b, a + 7b, ...

☐ 5a + 3, 3a + 4, a + 5, 6 - a, ...

☐ a, 2a + 4, 3a + 6, 4a + 8, ...

